

DANIEL LIU

Senior Software Engineer

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SUMMARY

Over **10 years** in software development, building scalable, secure, and high-performance applications for **fintech, healthcare**, eCommerce, and enterprise clients ranging from startup to large scale. Skilled in **Python, JavaScript, TypeScript** and **PHP**, with deep experience in microservices, cloud platforms, web/mobile frameworks, API development and integrating **AI** technologies. I excel at leading teams, implementing **DevOps** practices, **CI/CD** pipelines, and automated testing, driving efficiency and delivering solutions that accelerate growth and enhance user experiences. Passionate about transforming complex challenges into elegant solutions, I build scalable systems that empower businesses to thrive. Always coding with purpose and clarity, I deliver production-ready solutions that create measurable impact while adhering to a "**Quality over Quantity**" mindset in every line of code.

EXPERIENCE



Senior Software Engineer

Airbnb

04/2023 - Present

San Francisco, CA, US

- Architected and implemented cloud-native solutions on **Oracle Cloud Infrastructure**, utilizing **microservices**, **serverless** technologies, and event-driven architectures to build scalable, high-performance applications for multinational fintech and healthcare clients.
 - Built and optimized frontend applications using **React**, **Next.js**, and **TypeScript** for Oracle Cloud solutions, enhancing user experiences across multiple platforms by ensuring responsive, high-performance interfaces.
 - Implemented RESTful APIs with **Python**, **FastAPI**, and **Nest.js** integrated with **Oracle Cloud Infrastructure (OCI)** Object Storage, enabling secure data sharing across healthcare applications while maintaining HIPAA compliance.
 - Developed AI-powered healthcare solutions using **LangChain**, integrating **NLP models** for real-time extraction of medical insights from unstructured patient data, enabling faster decision-making for clinicians.
 - Engineered **RAG** pipelines with **LangChain**, combining retrieval-based data from **Oracle Autonomous Database** with **generative models (OpenAI, Hugging Face)** to automate patient outcome predictions and treatment recommendations, enhancing clinical decision support and improving healthcare delivery.
 - Designed event-driven serverless workflows for **Oracle Cloud Payments** using **OCI Functions**, **Apache Kafka**, and **FastAPI** to process millions of real-time transactions with 99.99% availability and PCI DSS compliance.
 - Led the redesign of **Oracle Fusion ERP financial modules** by developing scalable microservices with **Java**, **Spring Boot**, and **Oracle ADF** while optimizing **Oracle SQL** queries, achieving 32% faster transaction processing for enterprise clients.
 - Spearheaded **CI/CD** adoption with **Jenkins**, **GitHub Actions**, and **Oracle DevOps** by integrating automated testing frameworks such as **Jest**, **Cypress**, and **PyTest**, reducing release cycle time by 40% while improving deployment reliability.
 - Enhanced OCI's multi-cloud support by integrating **AWS S3** and **Azure Blob Storage** with **Terraform**, **Ansible**, and **Vault** for secret management, enabling hybrid cloud strategies that increased customer retention.
- Technologies:** Java, Python, TypeScript, React, Next.js, FastAPI, Spring Boot, LangChain, Oracle Cloud Infrastructure (OCI), Oracle SQL, Apache Kafka, Docker, Kubernetes, Jenkins, Terraform.



Teach Lead

UKG

09/2020 - 03/2023

Weston, Florida, US

- Introduced micro-frontend architecture to a multi-tenant SaaS product, leveraging **React**, **Next.js**, **TypeScript**, **Module Federation**, and **Docker**, reducing release bottlenecks and cutting delivery cycles.
- Engineered an AI-powered security analytics system for a government contractor, applying **Python**, **TensorFlow**, and **OpenCV** to deliver real-time video analysis of crowds, reducing incident response times.
- Implemented robust backend APIs using **Python**, **Flask**, and **Express.js** to handle complex business logic, user authentication, role-based access, and real-time data updates, improving system reliability and reducing response times, fully integrated backend APIs with **PostgreSQL** and **Redis** caching for high performance.

- Led development of a blockchain-based NFT marketplace for a fintech startup, using **Solidity**, **Web3.js**, and **Hardhat**, supporting over 120K token minting and exchange transactions within its first quarter.
- Built serverless backend microservices for a travel booking system using **AWS Lambda**, **API Gateway**, and **DynamoDB**, scaling automatically during seasonal traffic spikes, cutting infrastructure costs.
- Developed and integrated **CMS** and **CRM** solutions, customizing **WordPress** and **WooCommerce** for seamless **eCommerce** functionality, while connecting **CRM** systems to backend APIs for automated lead management and personalized marketing, improving both content management and customer engagement workflows.
- Implemented monitoring and alerting solutions using **Prometheus** for metrics collection and **Grafana** for real-time data visualization, ensuring system performance and reliability across microservices and reducing incident response times by 30%.
- Orchestrated containerized deployments across staging and production environments using **Docker** and **Kubernetes**, implementing **Helm charts** and **Istio** for service mesh, which improved reliability and reduced downtime during rollouts.

Technologies: React, Next.js, TypeScript, Python, Flask, Express.js, TensorFlow, Solidity, AWS Lambda, PostgreSQL, Redis, Docker, Kubernetes, Prometheus, Grafana.



Software Engineer

Paylocity

04/2018 - 08/2020

Schaumburg, IL, US

- Developed the Electronic Parts Catalog using **PHP**, **Laravel**, **MySQL**, and **jQuery**, indexing over 2M+ machine components with advanced filters, reducing part lookup time for manufacturers.
- Built an integrated B2B e-commerce solution with **Laravel**, **Stripe**, and **PayPal APIs**, supporting over \$10 million in annual aftermarket parts sales while ensuring PCI-DSS compliance.
- Optimized catalog performance through **Redis** caching and **MySQL** query tuning, cutting page load times from 4.2 seconds to 1.1 seconds, which improved **SEO rankings** and user engagement.
- Delivered a customer support portal with **Vue.js** and **REST APIs**, integrating case management with ERP workflows, reducing resolution times by 30% for equipment manufacturers.
- Implemented **CI/CD pipelines** with **Jenkins** and **GitHub Actions**, introducing automated **PHPUnit** and **Cypress** tests, which accelerated deployment frequency from monthly to bi-weekly.
- Developed a 3D product configurator tool with **JavaScript** and **Node.js** microservices, enabling customers to customize machinery builds online, which increased conversion rates by 22%.
- Partnered with UX designers to define persona-driven workflows in **Figma**, ensuring ease of adoption for non-technical manufacturing clients, resulting in a 38% increase in user adoption.

Technologies: PHP, Laravel, JavaScript, Vue.js, Node.js, MySQL, Redis, REST APIs, Stripe API, Jenkins, GitHub Actions, PHPUnit.



FullStack Developer

Appen

11/2015 - 03/2018

Sydney, NSW, Australia

- Contributed to the development of a mobile banking and lending app using **React** and **React Native**, integrating REST APIs with third-party payment gateways and implementing offline storage with **SQLite** for reliable access in low-connectivity regions.
- Built a healthcare appointment scheduling web platform using **React** and **Bootstrap**, connected to a **Node.js + Express** backend with **MongoDB**, and deployed on **Heroku**, helping clinics streamline patient bookings.
- Developed real-time logistics dashboards with **Socket.IO** and **Leaflet.js** maps, enabling fleet operators to track vehicles, visualize live telemetry data, and receive geofencing alerts directly in a browser interface.
- Integrated an AI-driven chatbot prototype into a customer support portal using **Dialogflow (API.AI at the time)** with a **Node.js** bridge, handling common queries and reducing first-response time by nearly 40%.
- Set up **CI/CD** pipelines with **Jenkins** and **Docker**, automating builds, testing, and deployments across staging environments, reducing deployment effort from several hours to less than 30 minutes.
- Translated **Sketch** and **Adobe XD** design prototypes into responsive, accessible web applications using **SASS**, **Bootstrap 4**, and **Vanilla.js**, ensuring cross-browser compatibility for enterprise users.
- Participated in **Agile Scrum** ceremonies, contributing to sprint planning, backlog refinement, and retrospectives, while mentoring junior developers on **JavaScript ES6** features, **REST API** design, and **Mocha** testing practices.

Technologies: React, React Native, Node.js, Express.js, MongoDB, SQLite, Socket.IO, Dialogflow, Docker, Jenkins, Heroku, Bootstrap.



- Developed internal admin dashboards using **AngularJS**, **jQuery**, and **Bootstrap 3**, enabling engineers to monitor chatbot interactions in near real-time.
- Built RESTful APIs with **Node.js** and **Express**, connected to **MongoDB** to fetch and update conversation logs, facilitating faster debugging of early NLP prototypes.
- Assisted in integrating Slack's Real Time Messaging API using **WebSockets**, implementing basic authentication flows with OAuth 2.0 for early chatbot messaging features.
- Created reusable UI components for the web console using **Handlebars.js**, standardizing the look and feel across different admin panels.
- Migrated small portions of legacy code from **jQuery**-heavy scripts to **AngularJS** directives, improving modularity and maintainability.
- Implemented **WebSocket** connections to stream live feedback from bot simulations to admin dashboards, enhancing visibility of message flows.
- Worked with junior data engineers to log chatbot telemetry in **MongoDB**, enabling early search and visualization of employee support queries.
- Wrote unit and integration tests using **Mocha** and **Chai**, achieving coverage for key API endpoints and dashboard components.

Technologies: AngularJS, JavaScript, Node.js, Express.js, MongoDB, REST APIs, WebSockets, OAuth 2.0, Mocha, Chai.

EDUCATION

University of Central Oklahoma

2010 - 2014

Bachelor of Science in Computer Science

SKILLS

- **Programming Languages:** JavaScript, TypeScript, Python, Java, PHP, Solidity, SQL, HTML5, CSS3, SASS
- **Frontend:** React, Next.js, Angular, Vue.js, Nuxt.js, React Native, Ionic, Redux, Vuex, Tailwind CSS, Bootstrap, jQuery
- **Backend:** Node.js, NestJS, Express.js, Laravel, Django, Flask, FastAPI, Spring Boot, Ruby on Rails
- **API & Communication:** GraphQL, WebSockets, Socket.IO
- **AI & Machine Learning:** TensorFlow, RAG (Retrieval-Augmented Generation), LangChain
- **Blockchain & Web3:** Solidity, Web3.js, Hardhat
- **Testing:** Jest, Mocha, Chai, PHPUnit, Cypress, Selenium
- **Cloud Platforms:** AWS, GCP, Azure, OCI, Firebase
- **Databases:** PostgreSQL, MongoDB, MySQL, Redis
- **DevOps & Infrastructure:** Docker, Kubernetes, Jenkins, Terraform, Ansible, HashiCorp Vault
- **Monitoring & Data Engineering:** Prometheus, Grafana, Apache Airflow, Apache Spark
- **Development Tools & Methodologies:** Git, GitHub, GitLab, Jira, Agile, Scrum, Kanban, Microservices Architecture

CERTIFICATIONS

- AWS Certified Solutions Architect – Professional | AWS | 2025
- Microsoft Certified: Azure AI Engineer Associate | Microsoft | 2024
- Certified Kubernetes Application Developer (CKAD) | CNCF | 2023
- HashiCorp Certified: Terraform Associate | HashiCorp | 2023